Tutorial 1: Probability and Multivariate Gaussians

Probabilistic Artificial Intelligence, HS 2026

Daniel Adorno & Lino Hofstetter

24 September 2026

Outline

- **Part I:** Probability recap
- **Part II:** Multivariate Gaussians
- **Part III:** Homework 1 tricks

Part I: Probability Recap

Joint and Conditional Distributions

Random vector $X = (X_1, \ldots, X_n)$ with joint distribution $P(X_1 = x_1, \ldots, X_n = x_n)$.

Sum rule (marginalization):

$$P(x_A) = \sum_{x_B} P(x_A, x_B) \qquad \left(\text{continuous: } p(x_A) = \int p(x_A, x_B)\, dx_B\right)$$

Conditional distribution ($P(x_B) > 0$):

$$P(x_A \mid x_B) = \frac{P(x_A, x_B)}{P(x_B)}$$

Product rule (chain rule):

$$P(x_1, \ldots, x_n) = P(x_1)\, P(x_2 \mid x_1) \cdots P(x_n \mid x_1, \ldots, x_{n-1})$$

Question 1

Joint distribution from the lecture:

	toothache		no toothache	
	catch	no catch	catch	no catch
cavity	.108	.012	.072	.008
no cavity	.016	.064	.144	.576

What is $P(\text{cavity} \mid \text{toothache})$?

(A) 0.12

(B) 0.2

(C) 0.6

(D) 0.8

Bayes' Rule

Product rule in both orders: $P(x, y) = P(y \mid x)\, P(x) = P(x \mid y)\, P(y)$. Hence

$$\underbrace{P(x \mid y)}_{\text{posterior}} = \frac{\overbrace{P(y \mid x)}^{\text{likelihood}}\ \overbrace{P(x)}^{\text{prior}}}{\underbrace{P(y)}_{\text{evidence}}}, \qquad P(y) = \sum_{x'} P(y \mid x')\, P(x')$$

In practice: posterior $\propto$ likelihood $\times$ prior. Normalize at the end.

Question 2

A test for a condition D:

$$P(D) = 0.01, \qquad P(+ \mid D) = 0.99, \qquad P(+ \mid \neg D) = 0.01$$

A person picked at random tests positive. What is $P(D \mid +)$?

(A) 0.99

(B) 0.5

(C) 0.01

(D) 0.0099

Independence and Conditional Independence

Independence $X \perp Y$:

$$P(x, y) = P(x)\,P(y) \;\; \forall x, y \quad \Longleftrightarrow \quad P(x \mid y) = P(x) \;\; (P(y) > 0)$$

Conditional independence $X \perp Y \mid Z$:

$$P(x, y \mid z) = P(x \mid z)\,P(y \mid z) \;\; \forall x, y, z \quad \Longleftrightarrow \quad P(x \mid y, z) = P(x \mid z)$$

Once Z is known, Y carries no further information about X.

Example (lecture): catch $\perp$ toothache $\mid$ cavity.

Question 3

Assume $X \perp Y \mid Z$. Which statement does **not** follow in general?

(A) $P(x \mid y, z) = P(x \mid z)$

(B) $P(x, y \mid z) = P(x \mid z)\, P(y \mid z)$

(C) $P(x, y, z) = P(z)\, P(x \mid z)\, P(y \mid z)$

(D) $P(x, y) = P(x)\, P(y)$

Question 4

X, Y are independent fair coins in $\{0, 1\}$ and $Z = X \oplus Y$ (XOR). Which is true?

(A) $X \perp Y$ and $X \perp Y \mid Z$

(B) $X \perp Y$, but not $X \perp Y \mid Z$

(C) $X \perp Y \mid Z$, but not $X \perp Y$

(D) Neither

Question 5

$Y, X_1, \ldots, X_n$ are binary and $X_1, \ldots, X_n$ are mutually independent given Y.

How many parameters specify $P(Y, X_1, \ldots, X_n)$?

(A) n

(B) $2n + 1$

(C) $2^n - 1$

(D) $2^{n+1} - 1$

Part II: Multivariate Gaussians

Multivariate Gaussian

$X \sim \mathcal{N}(\mu, \Sigma)$ with mean $\mu \in \mathbb{R}^d$ and covariance $\Sigma \in \mathbb{R}^{d \times d}$:

$$\mathcal{N}(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\Big(-\tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\Big)$$

$$\mu = \mathbb{E}[X], \qquad \Sigma = \mathbb{E}\big[(X-\mu)(X-\mu)^\top\big], \qquad \Sigma_{ij} = \mathrm{Cov}(X_i, X_j)$$

Valid covariance: $\Sigma = \Sigma^\top$ and, for all $a \in \mathbb{R}^d$,

$$a^\top \Sigma a = \mathrm{Var}(a^\top X) \geq 0.$$

The density requires $\Sigma \succ 0$. For 2×2: $\Sigma \succ 0 \iff \Sigma_{11} > 0$ and $\det \Sigma > 0$.

Parameters: $d + d(d+1)/2 = O(d^2)$.

Question 6

Which matrix is a valid covariance matrix of a Gaussian with a density?

(A) $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$

(B) $\begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$

(C) $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$

(D) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Question 7

How many parameters specify a Gaussian in $d = 10$ dimensions?

(A) 20

(B) 55

(C) 65

(D) 1023

Geometry of the Covariance Matrix

A **level set** of the density collects the points where it takes the same value. Since the density depends on x only through

$$\Delta^2(x) := (x-\mu)^\top \Sigma^{-1}(x-\mu),$$

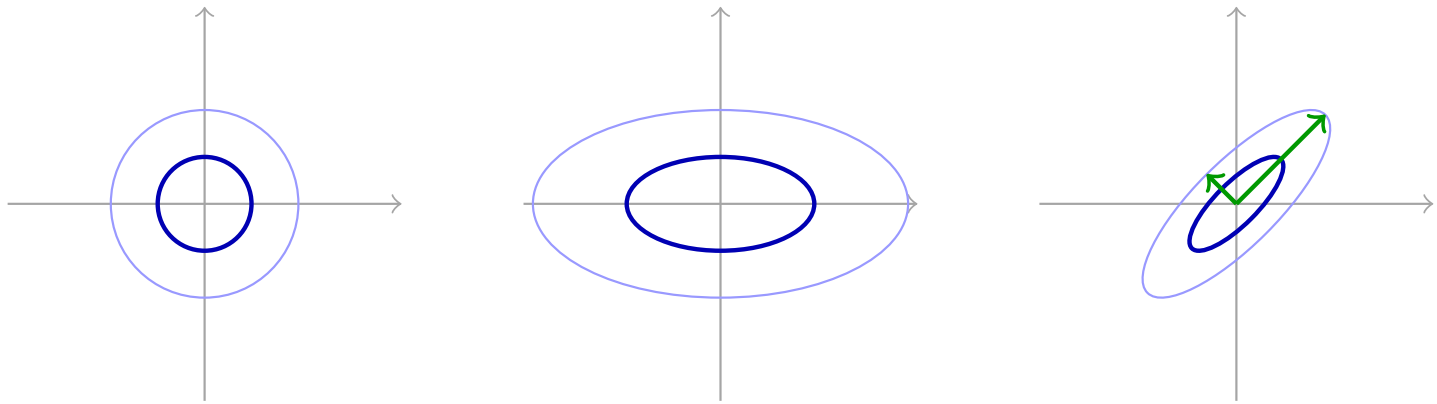
the level sets are $L_c = \{x : \Delta^2(x) = c\}$.

Spectral decomposition $\Sigma = UDU^\top$ with $U = (u_1, \ldots, u_d)$ orthogonal and $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$, $\lambda_i > 0$. Rotated coordinates $z = U^\top(x-\mu)$:

$$\Delta^2(x) = z^\top D^{-1} z = \sum_{i=1}^{d} \frac{z_i^2}{\lambda_i} = c \qquad \Rightarrow \qquad \sum_{i=1}^{d} \frac{z_i^2}{\left(\sqrt{c\lambda_i}\right)^2} = 1$$

- L_c is an ellipsoid: center μ, axes along u_i, half-lengths $\sqrt{c\,\lambda_i}$
- $\lambda_i = \mathrm{Var}(u_i^\top X)$: variance along u_i
- Δ counts standard deviations, one unit per direction

Geometry of the Covariance Matrix: Examples



$\Sigma = I$ $\quad$ $\Sigma = \mathrm{diag}(4, 1)$ $\quad$ $\Sigma = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}$

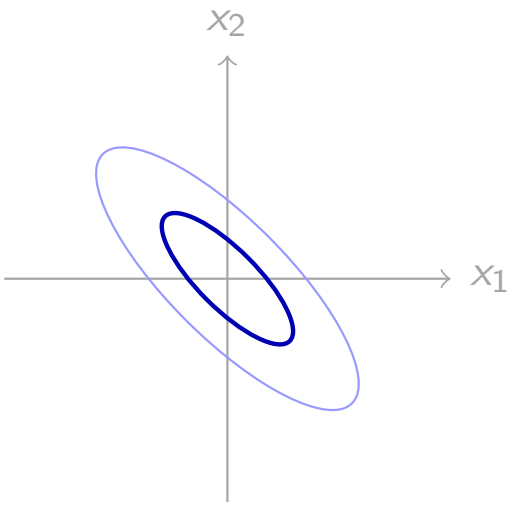
Contours at $\Delta^2 = 1$ and $\Delta^2 = 4$, that is at one and two standard deviations.

- $\lambda = 1, 1$: circle, same spread in every direction.
- $\lambda = 4, 1$: axes stay the coordinate axes, stretched by $\sqrt{\lambda_i} = 2$ and 1.
- $\lambda = 1.8,\ 0.2$: same shape, rotated to $u_1 = (1, 1)/\sqrt{2}$ (green arrows).

Rule: diagonal Σ gives axis-aligned ellipses, off-diagonal entries tilt them, the sign of Σ_{12} sets the tilt direction.

Question 8

The ellipses are level sets of $\mathcal{N}(0, \Sigma)$. Which Σ?



(A) $\begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}$

(B) $\begin{pmatrix} 1 & -0.8 \\ -0.8 & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 1.8 & 0 \\ 0 & 0.2 \end{pmatrix}$

(D) $\begin{pmatrix} 0.2 & 0 \\ 0 & 1.8 \end{pmatrix}$

Affine Transformations

Let $X \sim \mathcal{N}(\mu, \Sigma)$ in $\mathbb{R}^d$, $M \in \mathbb{R}^{m \times d}$, $b \in \mathbb{R}^m$ and $Y = MX + b$.

Mean (linearity of $\mathbb{E}$):

$$\mathbb{E}[Y] = M\,\mathbb{E}[X] + b = M\mu + b$$

Covariance (use $Y - \mathbb{E}[Y] = M(X - \mu)$):

$$\mathrm{Cov}(Y) = \mathbb{E}\big[M(X-\mu)(X-\mu)^\top M^\top\big] = M\Sigma M^\top$$

Gaussianity (proof via characteristic functions: HW 1, Ex. 2(c)):

$$Y \sim \mathcal{N}\big(M\mu + b,\ M\Sigma M^\top\big)$$

Special case $m = 1$: $a^\top X \sim \mathcal{N}(a^\top \mu,\ a^\top \Sigma a)$.

Question 9

$X \sim \mathcal{N}\left(0, \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\right)$. What is $\text{Var}(X_1 - X_2)$?

(A) 1

(B) 3

(C) 5

(D) 7

Three Consequences of the Affine Rule

1. Marginals. With $X = (X_A, X_B)$ and selection matrix $M = \begin{pmatrix} I & 0 \end{pmatrix}$:

$$X_A = MX \sim \mathcal{N}(\mu_A,\ \Sigma_{AA}) \qquad \text{(delete rows and columns of } B\text{)}$$

2. Independent sums. $X \perp X'$ gives $\begin{pmatrix} X \\ X' \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu \\ \mu' \end{pmatrix}, \begin{pmatrix} \Sigma & 0 \\ 0 & \Sigma' \end{pmatrix}\right)$.

With $M = \begin{pmatrix} I & I \end{pmatrix}$:

$$X + X' \sim \mathcal{N}(\mu + \mu',\ \Sigma + \Sigma')$$

Independence is needed: $X' = -X$ gives $X + X' = 0$.

3. Sampling. $Z \sim \mathcal{N}(0, I)$ and $LL^\top = \Sigma$:

$$\mu + LZ \sim \mathcal{N}(\mu,\ \Sigma)$$

Conversely, $L^{-1}(X - \mu) \sim \mathcal{N}(0, I)$ (whitening).

Question 10

$X \sim \mathcal{N}(\mu, \Sigma)$ with $\mu = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\Sigma = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & 2 & 5 \end{pmatrix}$. What is the distribution of (X_1, X_3)?

(A) $\mathcal{N}\left(\begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 & 1 \\ 1 & 5 \end{pmatrix}\right)$

(B) $\mathcal{N}\left(\begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 & 0 \\ 0 & 5 \end{pmatrix}\right)$

(C) $\mathcal{N}\left(\begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}\right)$

(D) $\mathcal{N}\left(\begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 & 2 \\ 2 & 5 \end{pmatrix}\right)$

Question 11

$X \sim \mathcal{N}(1, 2)$ and $Y \sim \mathcal{N}(-1, 3)$ are independent. What is the distribution of $X - 2Y$?

(A) $\mathcal{N}(3, -4)$

(B) $\mathcal{N}(3, 8)$

(C) $\mathcal{N}(3, 14)$

(D) $\mathcal{N}(-1, 14)$

Question 12

$Z \sim \mathcal{N}(0, I_2)$. Which X satisfies $X \sim \mathcal{N}\left(0, \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}\right)$?

(A) $X = \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix} Z$

(B) $X = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} Z$

(C) $X = \begin{pmatrix} 2 & 0 \\ 0 & \sqrt{2} \end{pmatrix} Z$

(D) $X = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} Z$

Uncorrelated $\Rightarrow$ Independent (Only for Jointly Gaussian)

Let $X = (X_A, X_B) \sim \mathcal{N}(\mu, \Sigma)$ with $\Sigma_{AB} = 0$. Then

$$\Sigma^{-1} = \begin{pmatrix} \Sigma_{AA}^{-1} & 0 \\ 0 & \Sigma_{BB}^{-1} \end{pmatrix}, \qquad \det \Sigma = \det \Sigma_{AA} \det \Sigma_{BB}.$$

The quadratic form splits:

$$(x-\mu)^\top \Sigma^{-1}(x-\mu) = (x_A-\mu_A)^\top \Sigma_{AA}^{-1}(x_A-\mu_A) + (x_B-\mu_B)^\top \Sigma_{BB}^{-1}(x_B-\mu_B)$$

Hence the density factorizes:

$$\mathcal{N}(x; \mu, \Sigma) = \mathcal{N}(x_A; \mu_A, \Sigma_{AA})\, \mathcal{N}(x_B; \mu_B, \Sigma_{BB}) \quad \Rightarrow \quad X_A \perp X_B$$

Independent $\Rightarrow$ uncorrelated holds for every distribution.

Requirement: (X_A, X_B) **jointly** Gaussian. Gaussian marginals are not enough.

Question 13

$X \sim \mathcal{N}(0,1)$, S uniform on $\{-1,+1\}$ and independent of X, $Y = SX$.

Which statement is **false**?

(A) $Y \sim \mathcal{N}(0,1)$

(B) $\mathrm{Cov}(X,Y) = 0$

(C) X and Y are independent

(D) (X,Y) is not jointly Gaussian

Question 14

$X \sim \mathcal{N}\left(0, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right)$ with $\rho \neq 0$. For which c are X_1 and $X_2 - cX_1$ independent?

(A) $c = 0$

(B) $c = \rho$

(C) $c = 1/\rho$

(D) $c = -\rho$

Conditioning: Derivation

$X = (X_A, X_B) \sim \mathcal{N}(\mu, \Sigma)$. Goal: law of X_A given $X_B = x_B$.

Split X_A into a part explained by X_B and a residual:

$$E := X_A - \Sigma_{AB}\Sigma_{BB}^{-1}X_B$$

1. (E, X_B) is an affine map of X, hence jointly Gaussian.
2. $\mathrm{Cov}(E, X_B) = \Sigma_{AB} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BB} = 0$, hence $E \perp X_B$.
3. Mean and covariance of the residual:

$$\begin{aligned}
\mathbb{E}[E] &= \mu_A - \Sigma_{AB}\Sigma_{BB}^{-1}\mu_B \\
\mathrm{Cov}(E) &= \Sigma_{AA} - 2\,\Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA} + \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BB}\Sigma_{BB}^{-1}\Sigma_{BA} \\
&= \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}
\end{aligned}$$

Conditioning: Result

Rewrite

$$X_A = \Sigma_{AB}\Sigma_{BB}^{-1}X_B + E, \qquad E \perp X_B.$$

Fix $X_B = x_B$. The first term becomes a constant. E keeps its law because $E \perp X_B$:

$$X_A \mid X_B = x_B \sim \mathcal{N}\big(\Sigma_{AB}\Sigma_{BB}^{-1}x_B + \mathbb{E}[E],\ \mathrm{Cov}(E)\big)$$

$$\boxed{\mu_{A|B} = \mu_A + \Sigma_{AB}\Sigma_{BB}^{-1}(x_B - \mu_B), \qquad \Sigma_{A|B} = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}}$$

HW 1, Ex. 2(b) derives the same result from the density.

Question 15

$(X, Y) \sim \mathcal{N}\left(0, \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}\right)$. We observe $Y = 1$.

What is the law of $X \mid Y = 1$?

(A) $\mathcal{N}(1, 1)$

(B) $\mathcal{N}(1, 2)$

(C) $\mathcal{N}(0.5, 1)$

(D) $\mathcal{N}(0, 2)$

Gaussian Cheat Sheet

$X \sim \mathcal{N}(\mu, \Sigma)$ partitioned into blocks A, B. Precision $\Lambda = \Sigma^{-1}$.

Affine map	$MX + b \sim \mathcal{N}(M\mu + b,\ M\Sigma M^\top)$
Marginal	$X_A \sim \mathcal{N}(\mu_A,\ \Sigma_{AA})$
Conditional mean	$\mu_{A\|B} = \mu_A + \Sigma_{AB}\Sigma_{BB}^{-1}(x_B - \mu_B)$
Conditional covariance	$\Sigma_{A\|B} = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA} = (\Lambda_{AA})^{-1}$
Independent sum	$X + X' \sim \mathcal{N}(\mu + \mu',\ \Sigma + \Sigma')$
Independence	$X_A \perp X_B \iff \Sigma_{AB} = 0$
Cond. independence	$X_i \perp X_j \mid X_{\text{rest}} \iff \Lambda_{ij} = 0$

Part III: Homework 1, Tricks and Tactics

Homework 1: Which Tool Where

Exercise	Tool
0(a)	sum rule, product rule
0(b), 0(c)	completing the square
1(a), 1(b)	affine maps, zero covariance, conditioning trick
1(c)	parameter counting, spectral theorem
2(a), 2(b)	completing the square in blocks, Schur complement
2(c), 2(d)	characteristic functions
2(e) to 2(g)	affine maps, conditioning formula, gradients
3	Bayes' rule, conjugate priors, MAP and MLE
4	running sums, Sherman–Morrison, affine maps

Ex. 2(g) and 4 use the Bayesian linear regression model (lecture 2). The model is stated in the exercise. Today's tools suffice.